RIYADH HEATREADY

Urban growth and surface heat screening

A priority map built from satellite data for southeast Riyadh

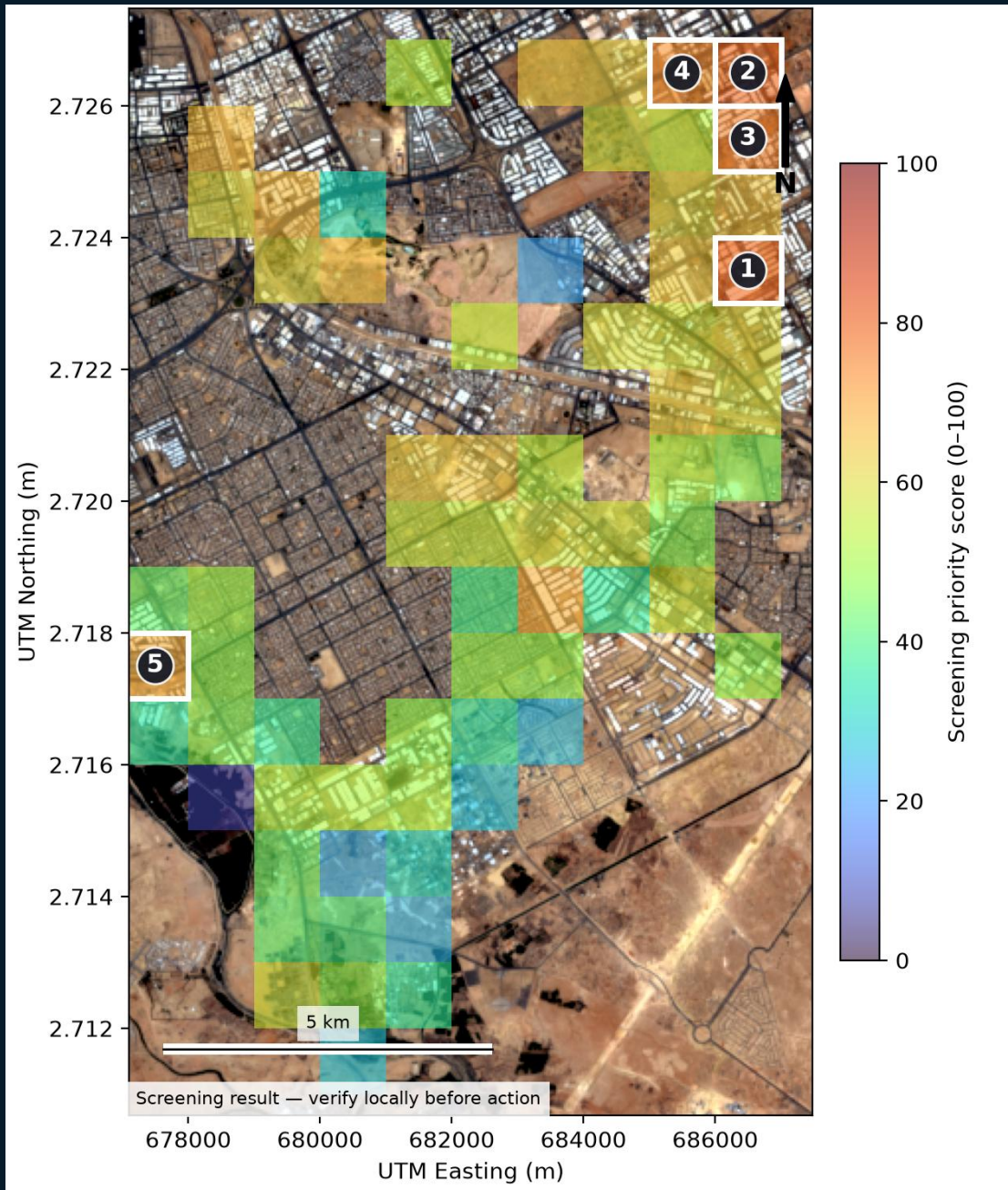
THE QUESTION

Where should a municipal planning team inspect first?

Team Milkaholics

Yusuf Karanib and Raed Mardoum

PRIORITY MAP



Our team

Yusuf Karanib

Project lead and demonstration

Leads the final submission. Explains how the satellite data becomes a priority map and runs the demonstration.

Raed Mardoum

Validation and pilot

Explains what the validation results mean and where the method is limited. Covers how planners could test the ranking in a municipal pilot.

A faster way to choose areas for inspection

168.30 km²
SOUTHEASTERN RIYADH STUDY AREA

1 km
TRANSPARENT SCREENING CELLS

1 user
MUNICIPAL PLANNING + HEAT TEAM

1 action
INSPECT LOCALLY BEFORE INTERVENTION

PROBLEM

Growth, heat, population and vegetation sit in separate layers. A planner still needs a short, explainable list of places to inspect.

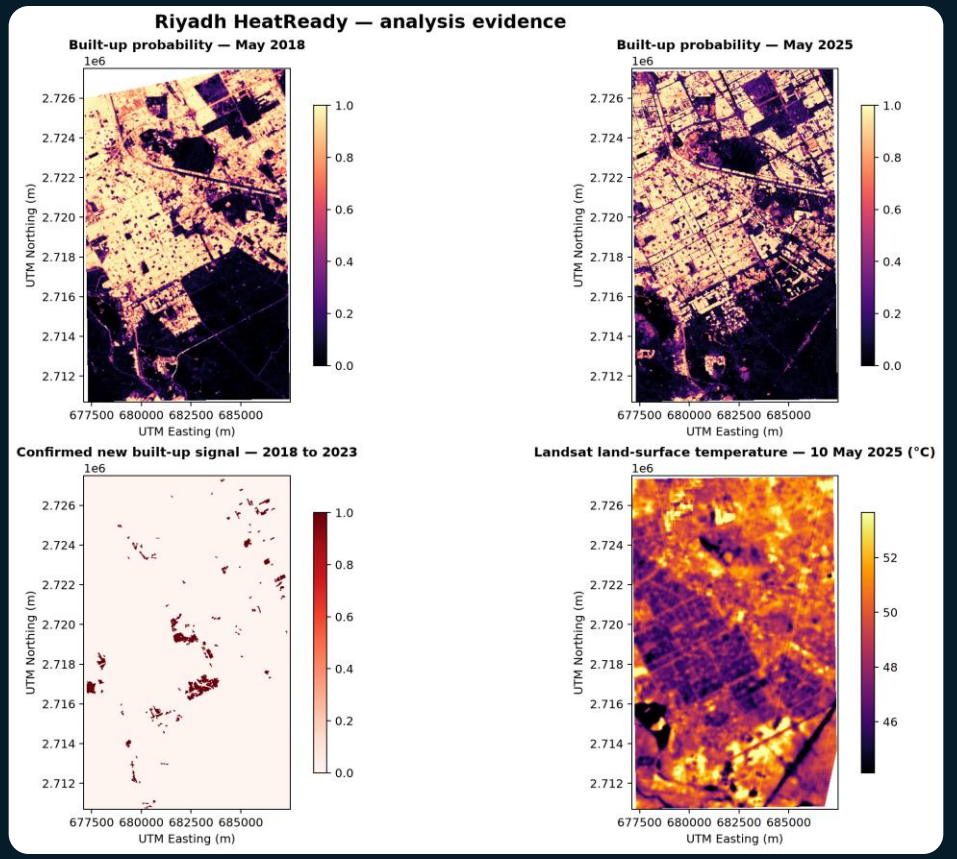
OUTPUT

HeatReady combines the layers into a ranked list of 1 km cells for local review. Local checks come before any intervention.

Current limitation: we have not interviewed municipal users. A pilot must test whether this fits their workflow.

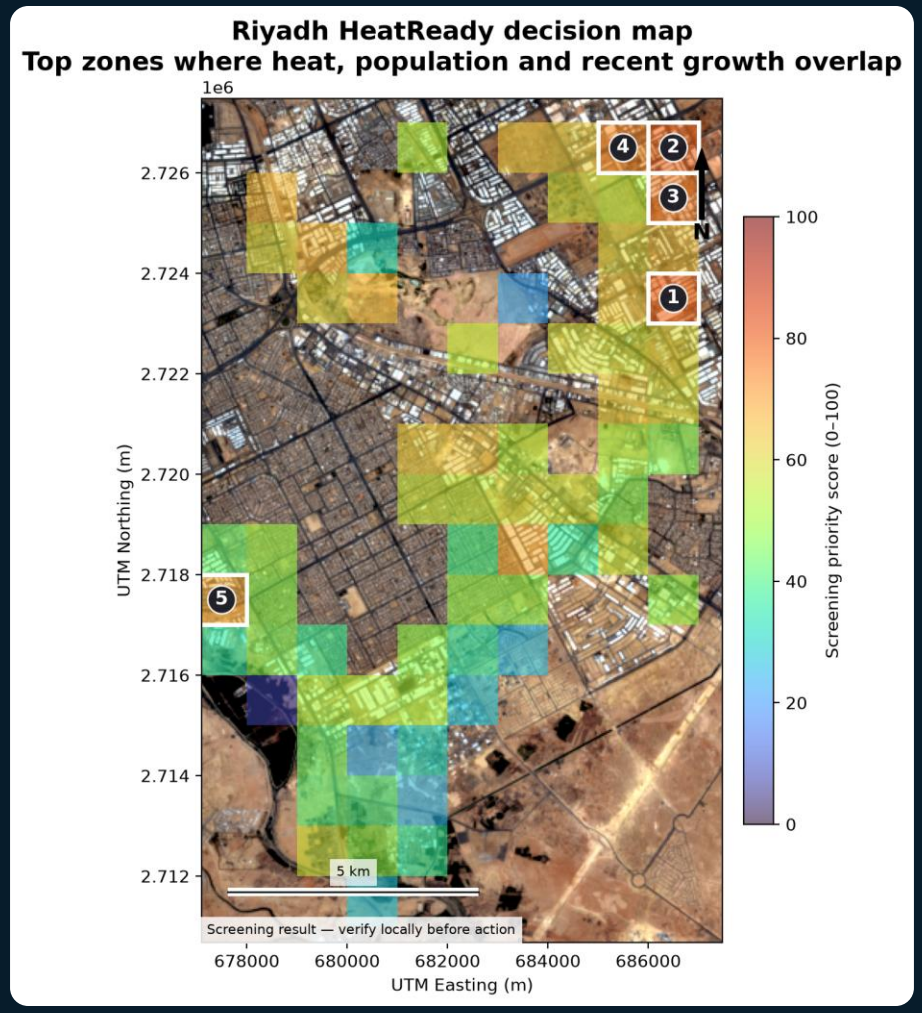
Data used in the analysis

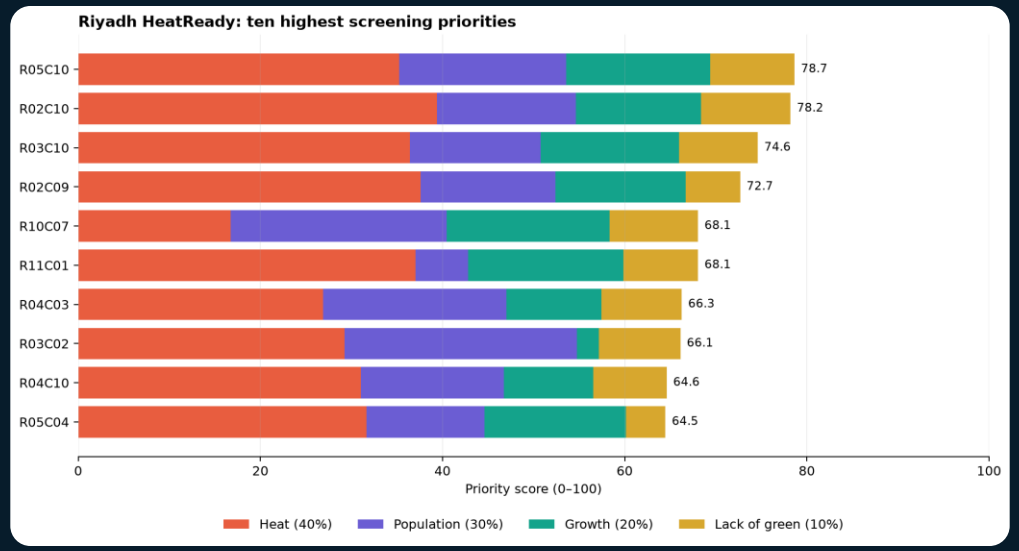
Sentinel-2	Optical reflectance, vegetation and built features
Sentinel-1	Radar structure to reduce confusion in bright desert areas
Annual LULC	Six maps propose 2018–2023 built-up change
Landsat 8	2025 land surface temperature after quality masking
WorldPop	Modelled 2025 population exposure
Planet Tanager	426-band context and a test of a known desert error



Annual maps flag possible growth. Sentinel-1 and Sentinel-2 check whether it looks built-up. Heat, population and vegetation then produce the ranking.

Priority map and GIS files





3.74 km²
CONFIRMED GROWTH SIGNAL

14 / 67
HIGH / ELIGIBLE CELLS

78.7
TOP SCORE: R05C10

The HTML report shows the map and a table of the top five zones. Separate CSV, GeoJSON and GeoTIFF files support further review.

Validation results and main limitations

0.88

F1 AGREEMENT

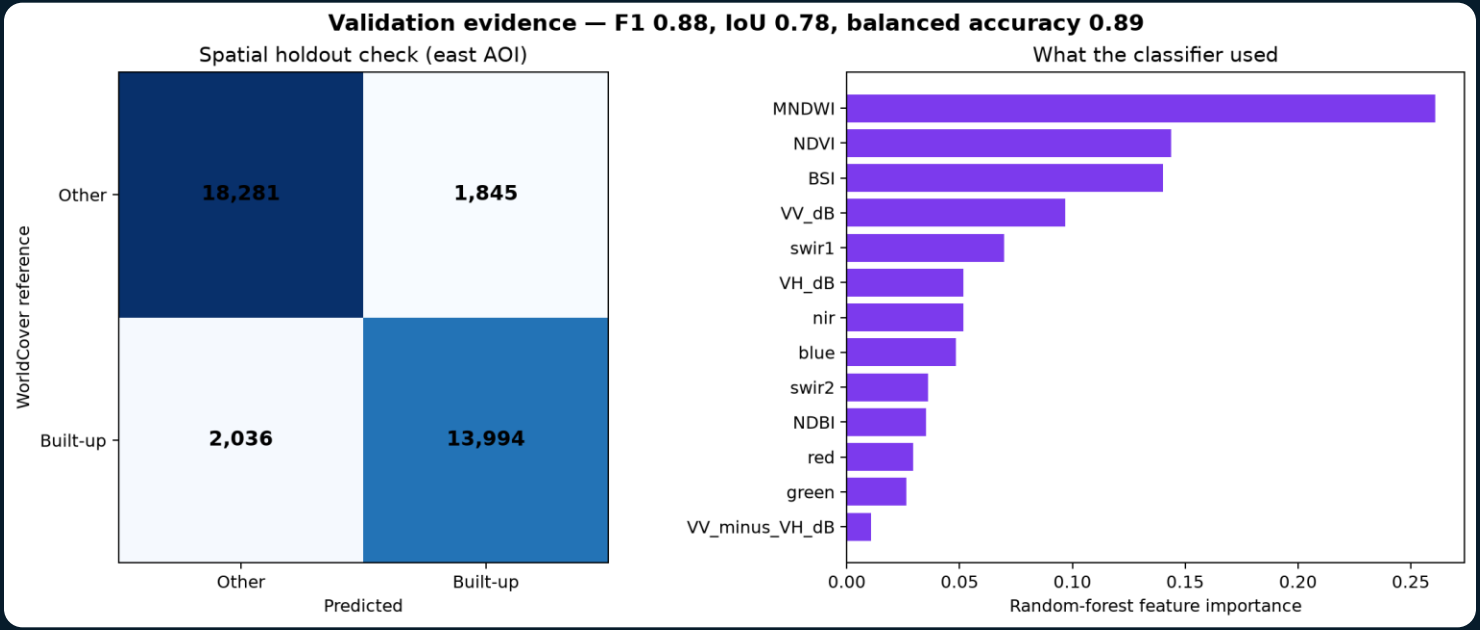
0.78

INTERSECTION OVER UNION

0.89

BALANCED ACCURACY

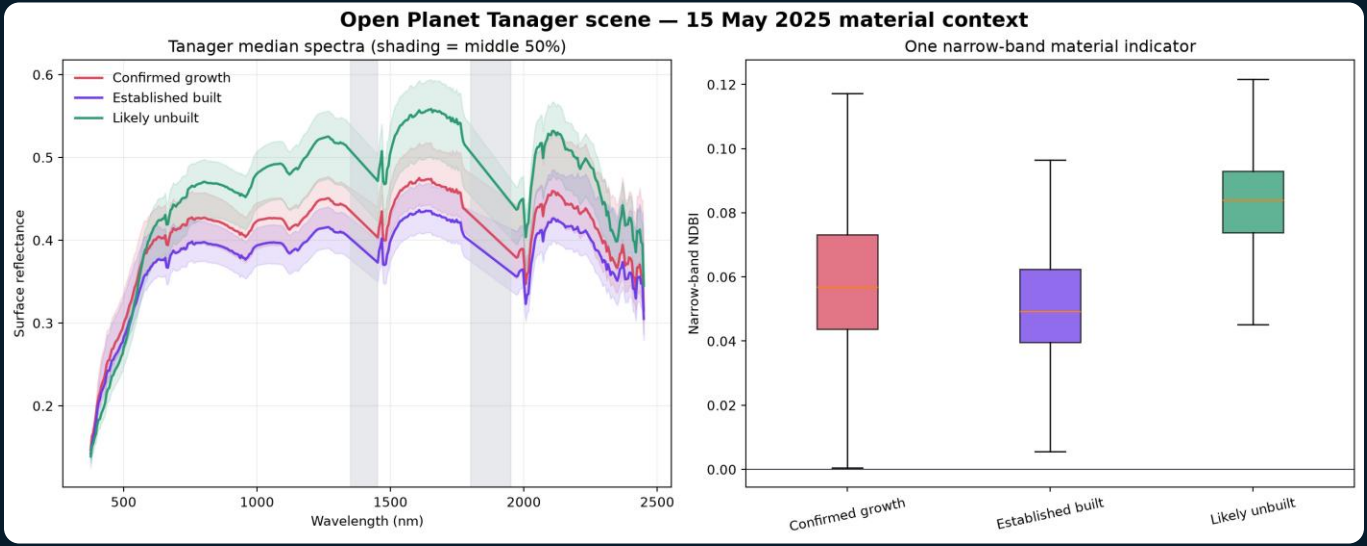
These numbers show agreement with a separate WorldCover reference. Field checks are still needed.



MAIN LIMITATIONS

- Land surface temperature cannot measure air temperature or personal heat exposure.
- WorldPop estimates population from a model. It does not provide a live census or worker count.
- Bright desert and roofs can still confuse classification.
- Score weights are PoC assumptions that a real user must review.

Evidence from several satellite sources



METHOD

- Annual land cover maps flag possible growth.
- Optical and radar images check whether the area still looks built-up.
- The output shows every part of the score and includes the GIS evidence.

One Tanager scene adds spectral context for a single date. It cannot prove change or identify materials.

POSSIBLE BENEFIT

A planner can review one ranked list instead of comparing several separate layers. This still needs user testing.

Proposed municipal pilot

1

Interview

Confirm the decision, current workflow and acceptable false alarms with at least three real users.

2

Field check

Inspect the top five zones, record useful results and false alarms, and compare with local plans.

3

Adjust

Agree on weights, add worker exposure and city-plan constraints, then rerun the ranking.

4

Repeat

Test the same process in another Riyadh area before making an operational claim.

PILOT REQUEST

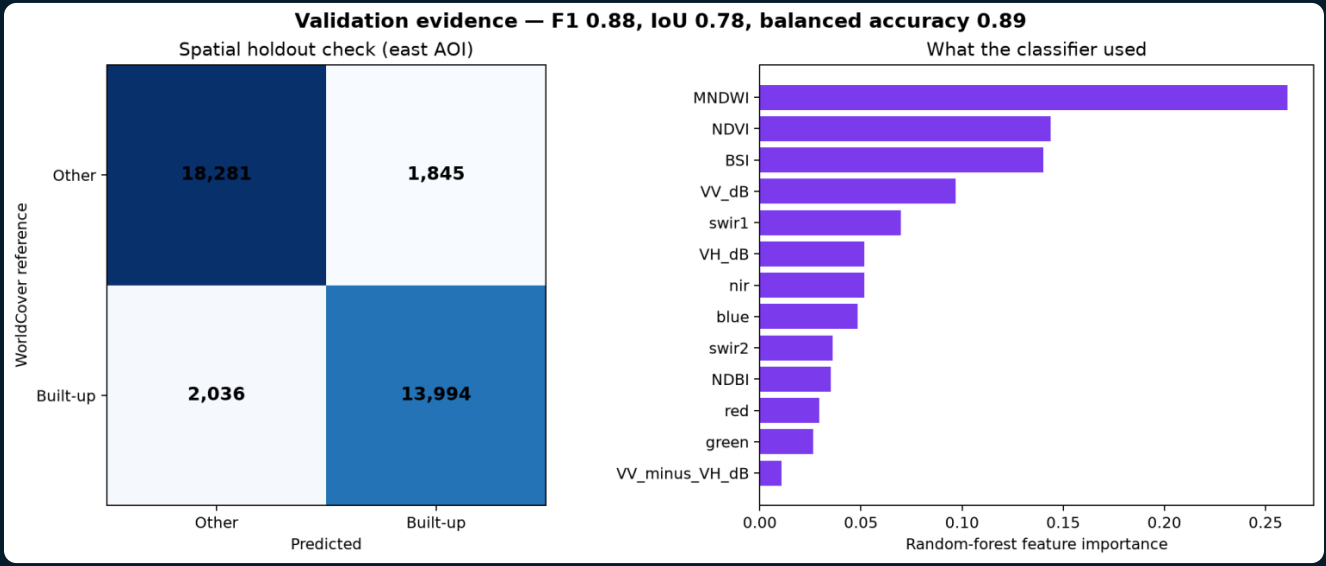
Test HeatReady with one municipal team and measure useful findings, false alarms and time spent.

Data sources and dates

Source	Dates used	Role	Access / licence
Sentinel-2 L2A	11 May 2018; 10 May 2021; 9 May 2025	Optical features, masks, vegetation	Copernicus open access
Sentinel-1 RTC	10 May 2018; 6 May 2021; 9 May 2025	Radar structure	Microsoft PC; CC BY 4.0
IO/Esri annual LULC	2018–2023	Trend + candidate endpoint change	CC BY 4.0
Landsat 8 C2 L2	10 May 2025	Land-surface temperature	US public domain
ESA WorldCover	2021	Spatial holdout reference	CC BY 4.0
WorldPop Global2	2025	Modelled population	CC BY 4.0
Planet Tanager	15 May 2025	Spectral context + sensitivity test	Open urban archive; CC BY 4.0

Item IDs, processing steps and licences are documented in the repository. Satellite 813 imagery was not used.

Validation evidence and limits



Training area: west Tuning area: middle Test area: east

RESULTS

- The east-side holdout was compared with WorldCover.
- F1 0.88, IoU 0.78 and balanced accuracy 0.89.
- The test contains 36,156 reference pixels.

LIMITS

- Field accuracy.
- Air temperature or health outcomes.
- Municipal approval or user acceptance.

Next: field and user review of the top-ranked cells.